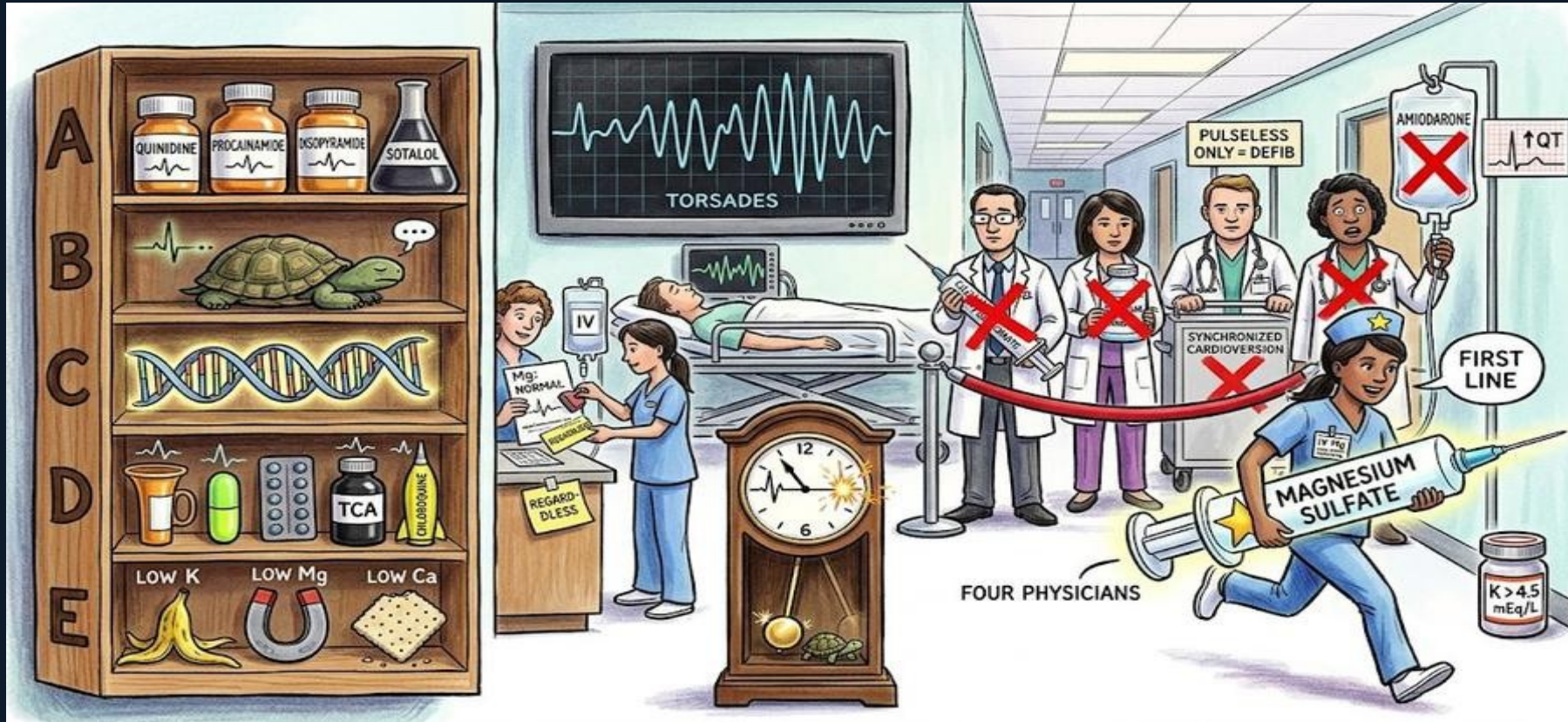
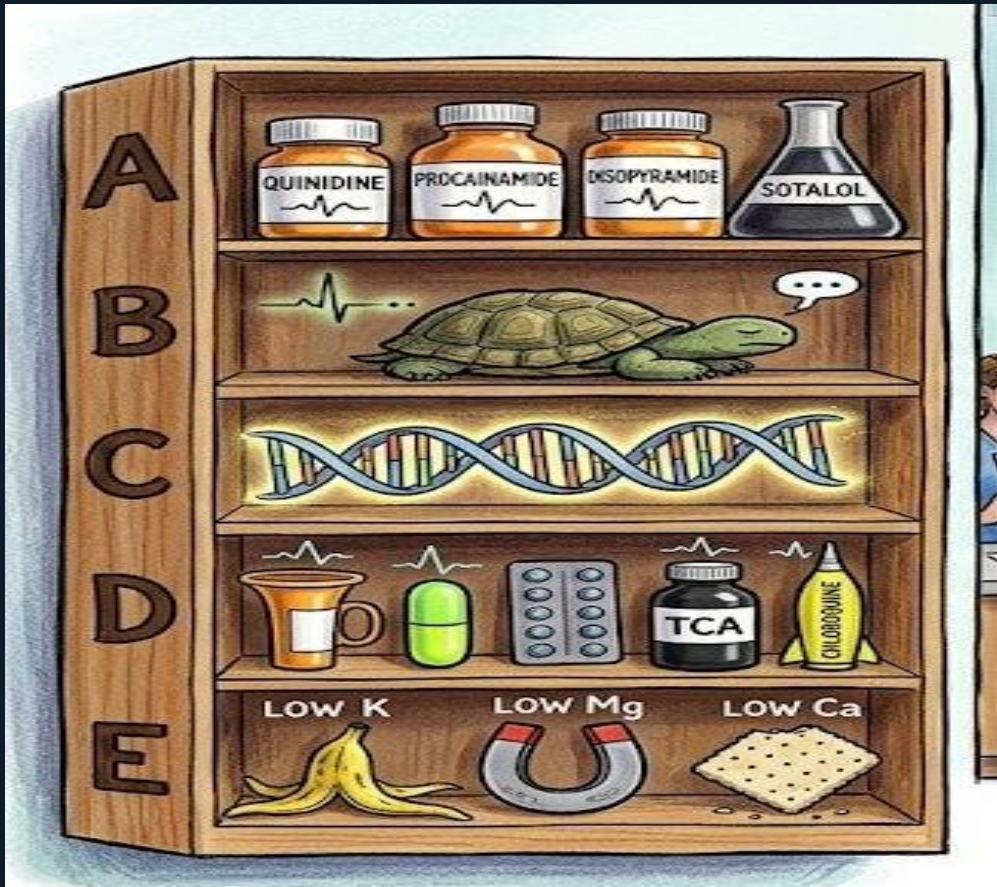


Scene A: Torsades de Pointes & Long QT



ER Triage Corridor • MUST MEMORIZE • 4 Board Questions



The ABCDE Shelf

A wooden shelf labeled A through E holds the five categories of long QT causes — the mnemonic framework the boards expect you to apply instantly when you see torsades.

Each shelf row = one category. The moment you see torsades on an ECG, your brain should run through A-B-C-D-E to identify the cause and choose the right treatment.

Board approach: a question giving you a patient with polymorphic VT and a prolonged QTc will expect you to identify the cause from this framework before selecting treatment.

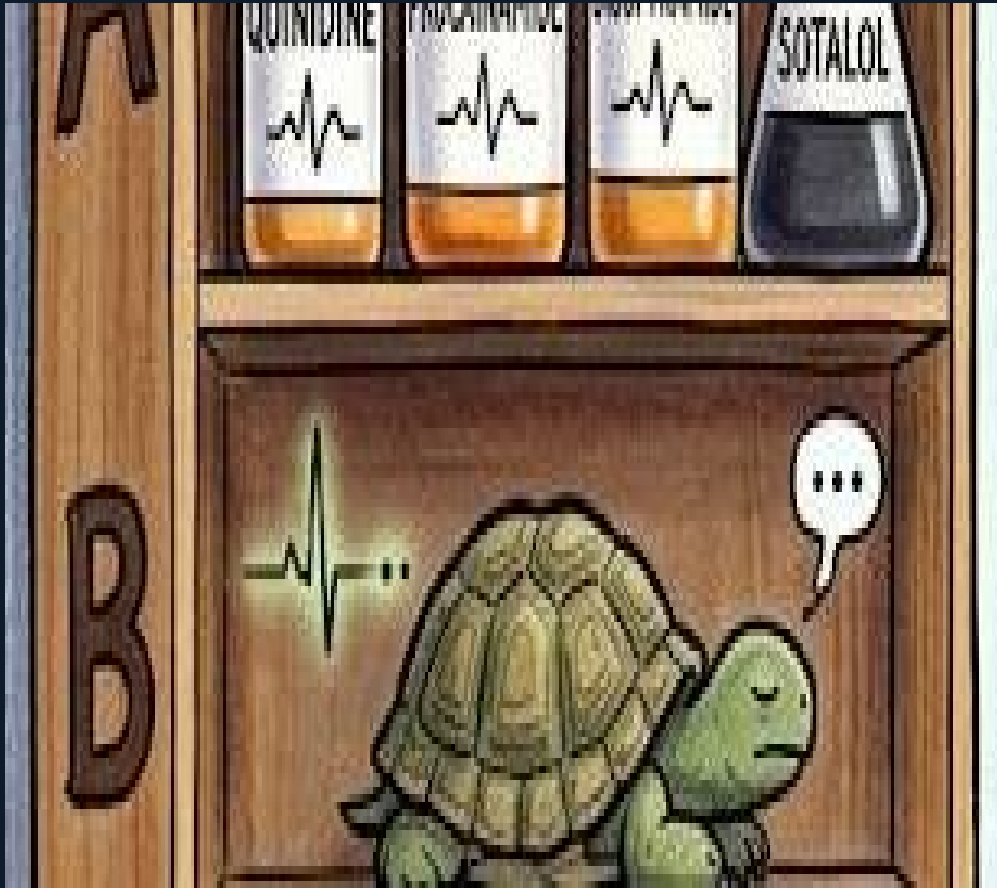


Shelf A — Antiarrhythmics

Three bottles — QUINIDINE, PROCAINAMIDE, DISOPYRAMIDE — stand beside a SOTALOL flask. All four have a QT-prolonging wave on their labels.

These are the antiarrhythmic drugs that cause torsades: class IA (quinidine, procainamide, disopyramide) and class III (sotalol, dofetilide, ibutilide). They prolong the action potential duration → extend QT.

Board trap: a patient started on procainamide or quinidine who develops polymorphic VT — the drug itself is the cause. Stop it immediately. Do not add amiodarone.

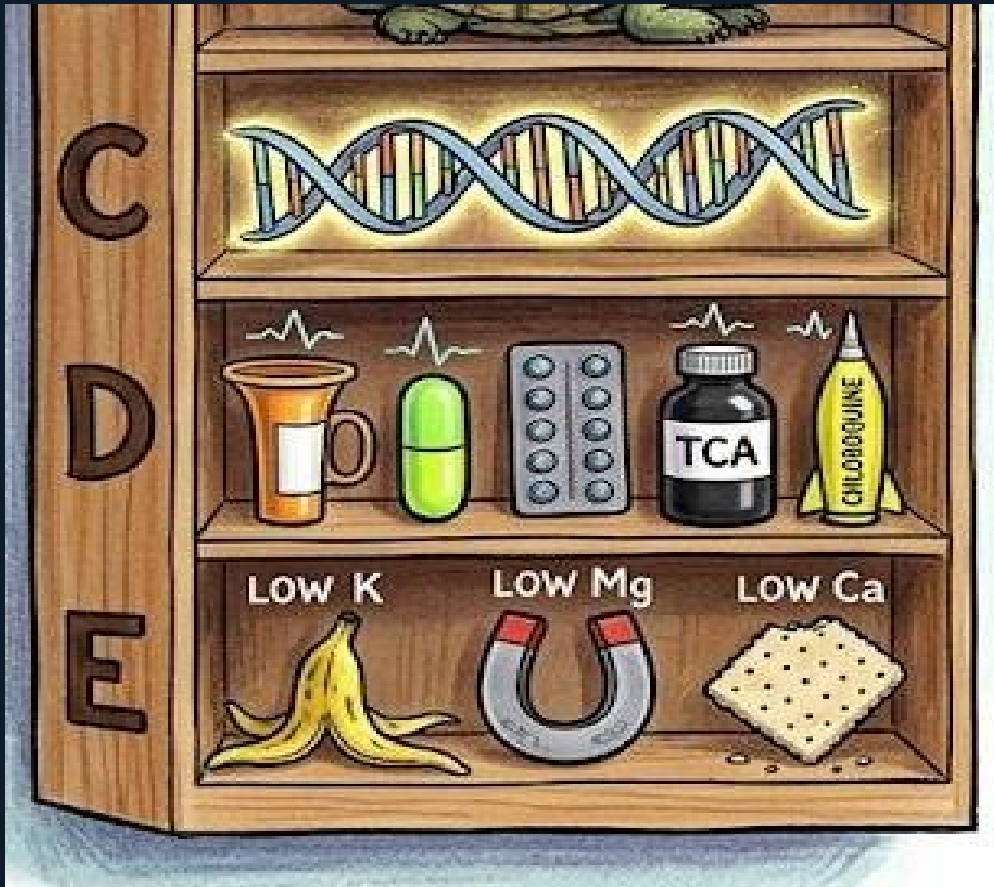


Shelf B — Bradycardia / Pause-Dependent

A sleeping tortoise with a slow heartbeat blip sits on shelf B — bradycardia as a cause of long QT.

Torsades is classically 'pause-dependent': a long pause between beats allows early afterdepolarizations (EADs) to reach threshold and trigger the arrhythmia. This is why torsades often follows a compensatory pause after a PVC.

Board trap: the pause clock in the center of the scene encodes this mechanism — the pendulum freezes, sparks erupt during the long pause. Treatment: overdrive pacing or isoproterenol to eliminate the pause.

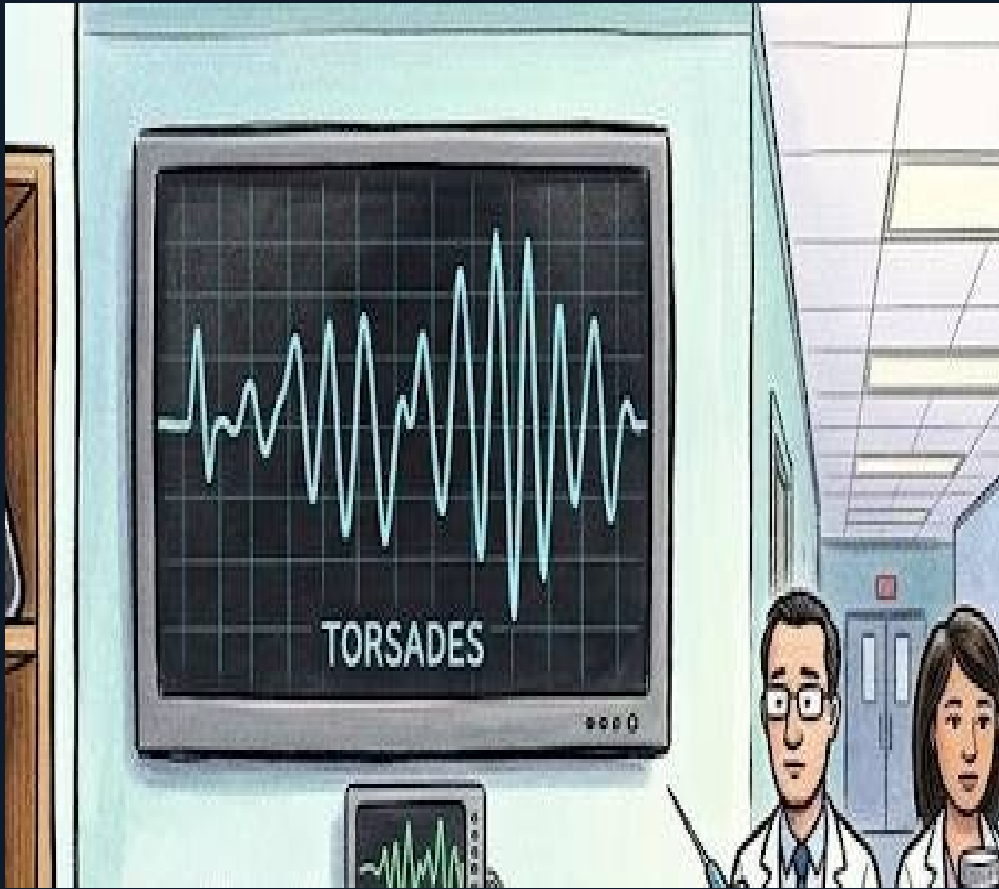


Shelves C, D, E — Congenital, Drugs, Electrolytes

Shelf C: glowing DNA helix = congenital long QT syndromes (Romano-Ward, Jervell-Lange-Nielsen).

Shelf D: five drug bottles — macrolide antibiotic, fluoroquinolone, antipsychotic, TCA, and CHLOROQUINE dart. All extend the QT. Shelf E: depleted banana peel (LOW K), faded magnet (LOW Mg), crumbling cracker (LOW Ca).

Board trap: chloroquine on shelf D is a tested drug. Any question with an antimalarial + polymorphic VT = torsades from QT prolongation. Electrolyte deficiency (especially hypokalemia and hypomagnesemia) potentiates QT prolongation from any other cause.



Scene A — Torsades de Pointes / Long QT | Arrhythmias Board Prep

The Torsades ECG

The monitor shows a polymorphic waveform that twists and spirals around the baseline — the characteristic 'twisting of the points' that names torsades de pointes.

This distinguishes torsades from monomorphic VT (same QRS morphology every beat) and from VF (completely chaotic). Torsades: polymorphic + baseline QTc prolonged on the preceding rhythm strip.

Board trap: 'synchronized cardioversion for torsades' — WRONG. If pulseless, this is unsynchronized defibrillation. If pulsed and stable, the treatment is IV magnesium — not cardioversion.



The Pause Clock

The grandfather clock has a frozen pendulum. On its face: two heartbeat spikes, then a long flat pause line, then sparks erupting — the afterdepolarizations that trigger torsades.

The tortoise inside the pendulum compartment reinforces bradycardia as the setting. This pause-dependent mechanism explains why torsades occurs after PVCs (compensatory pause), in bradycardia, and after long RR intervals.

Clinical key: isoproterenol and overdrive pacing both work by eliminating the pause — they increase the heart rate so the long RR interval never occurs.



Scene A — Torsades de Pointes / Long QT | Arrhythmias Board Prep

Normal Mg — Give It Anyway

A nurse holds a lab result reading 'Mg: NORMAL' and stamps it with a sticky note: 'REGARDLESS.' She simultaneously hands an IV magnesium bag to a colleague.

This encodes the highest-yield torsades board fact: IV magnesium is first-line even when serum magnesium is normal. The mechanism is not correction of deficiency — magnesium stabilizes the cardiac membrane and suppresses EADs regardless of serum level.

Board trap: 'serum Mg is normal so magnesium is not indicated' — WRONG. Give IV magnesium sulfate regardless of serum level.



Scene A — Torsades de Pointes / Long QT | Arrhythmias Board Prep

The Three Blocked Wrong Answers

Three physicians are stopped by a red velvet rope.

Physician 1: CALCIUM GLUCONATE syringe — red X.

Physician 2: SODIUM BICARBONATE vial — red X.

Physician 3: SYNCHRONIZED CARDIOVERSION cart — red X with 'PULSELESS ONLY = DEFIB.'

Calcium gluconate: correct for hyperkalemia-induced cardiac toxicity, not torsades. Sodium bicarbonate: correct for sodium channel blocker toxicity (TCA overdose), not torsades. Synchronized cardioversion: correct for stable tachyarrhythmia with pulse — but torsades with pulse gets IV magnesium first.

Board shortcut: torsades treatment = IV Mg → correct electrolytes → stop offending drug.



Scene A — Torsades de Pointes / Long QT | Arrhythmias Board Prep

Amiodarone — Red X on the Bag

A fourth physician holds an IV bag labeled 'AMIODARONE' with a large red X stamped directly on it. Beside the bag, an ECG strip shows a QT interval with an upward arrow: '↑QT.'

Amiodarone is a class III antiarrhythmic that prolongs the QT interval — exactly what caused the torsades in the first place. Giving amiodarone in torsades risks worsening the arrhythmia or re-triggering it after cardioversion.

Board trap: 'amiodarone for VT' — correct for monomorphic VT. For torsades (polymorphic VT + long QT), amiodarone is contraindicated. The QT distinction is the key.



IV Magnesium — First Line

The heroic nurse sprints forward with an enormous glowing syringe labeled 'MAGNESIUM SULFATE.' A gold star is pinned to the syringe. Her badge: 'IV Mg — EVEN NORMAL SERUM Mg.' Speech bubble: 'FIRST LINE.'

IV magnesium sulfate 2g IV over 5–15 minutes is the immediate treatment for torsades. It suppresses EADs and terminates the arrhythmia without requiring serum magnesium deficiency.

After magnesium: correct potassium (target $K > 4.5$), stop all QT-prolonging drugs, address the underlying bradycardia/pause if present. The $K > 4.5$ jar in the corner encodes this second step.

Scene A — Board Trap Master Summary

ABCDE = Long QT causes

Antiarrhythmics, Bradycardia, Congenital, Drugs, Electrolytes

Class IA → torsades

Quinidine, procainamide, disopyramide prolong QT

Pause-dependent mechanism

Long RR → EADs → torsades — treat with pacing/isoproterenol

Sodium bicarb = WRONG

For TCA/Na-channel blocker toxicity, not torsades

Amiodarone = WRONG

↑ QT → worsens torsades; use only for monomorphic VT

IV Mg even if serum normal

Membrane stabilization, not deficiency correction

Chloroquine prolongs QT

Antimalarial on shelf D — board-tested drug

Calcium gluconate = WRONG

For hyperkalemia, not torsades

Synch cardioversion = WRONG

Torsades + pulse → IV Mg; pulseless → defibrillation

K target >4.5 mEq/L

Repleting K reduces QT dispersion in torsades